

How do different brewing conditions (pH) affect the catechin content in green tea, as determined by titration with potassium permanganate?

Chemistry HL Internal Assessment

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Introduction

Green tea is a drink beyond the act of drinking in my culture— it is rich in traditions and heritage wisdom, a sign of hospitality and good health. Used as medicine in China for generations, tea remains one of the most popular drinks consumed today, said to have numerous related health benefits such as slowing the process of aging and boosting immunity.

These beliefs have been proven by scientific research with the discovery of polyphenols in green tea. Polyphenols are plant chemicals that are rich in plants, categorized depending on their chemical structure of one or more rings of benzene attached to a single or several hydroxyl groups (-OH). My study will focus on Catechin, a type of polyphenol that is particularly abundant and most active in green tea (Fan et al.). Catechins have multiple hydroxyl (-OH) groups on their benzene rings (See Figure 1). This means it can easily donate electrons or hydrogen atoms, which makes it a strong reducing agent or antioxidant (Tsao). This is particularly important because aging, inflammation and various diseases in the human body are caused by oxidative stress. Oxidative stress occurs when there is an imbalance between free radicals (reactive oxygen species) and antioxidants in the body (Pizzino et al.). Free radicals' free electrons are highly reactive, and when not regulated, they can do harm to cells, proteins, lipids, and DNA. Therefore, catechins have the ability to neutralize free radicals by donating electrons to protect cells from damage (Harvard Health Publishing). This protective mechanism also plays a crucial role in inhibiting tumor growth, and preventing cardiovascular diseases (Bernatoniene and Kopustinskiene). Apart from these health benefits, catechins also significantly influence the taste of tea, contributing primarily to its bitterness and astringency.

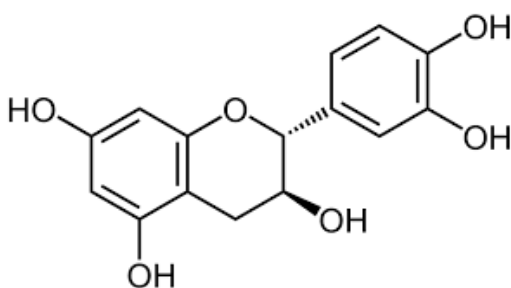


Figure 1 Catechin structure (Chattopadhyay et al.)

In my culture, a great deal of significance is placed on the method of brewing, ranging from water temperature, type of water, to method of brewing, which all have an effect on the flavor and health impacts of tea. There is a belief that slightly acidic spring water is best for brewing, enhancing the healing potential of tea. This interested me—does pH actually influence the chemical composition of tea, or is this all beliefs rooted in traditions rather than science? To

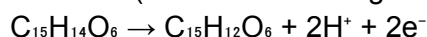
explore this, I decided to examine the impact that pH has on the concentration of catechin in green tea. Since catechins in green tea possess a high reducing property, a redox titration can be used in an effort to examine the catechin yield in water at various levels of pH.

For the experiment, I will use potassium permanganate titration, a common redox titration method. Potassium permanganate (KMnO_4), being a strong oxidizing agent titration is used as the titrant to react with catechins in an acidic medium.

Redox Reaction in Acidic Medium (H_2SO_4)

The balanced half-reactions are:

1. Oxidation (Catechin Donating Electrons):



2. Reduction (KMnO_4 Accepting Electrons):



Overall Reaction:

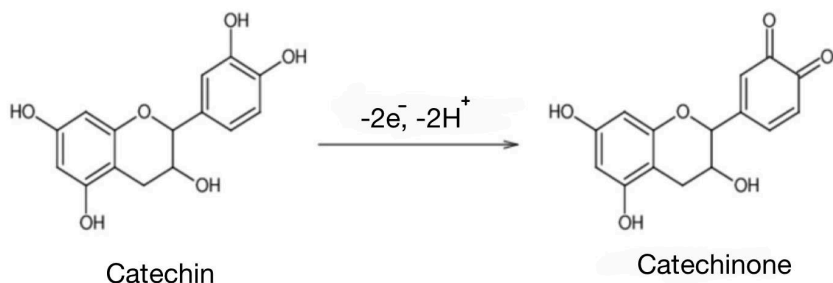
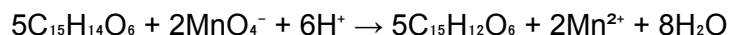
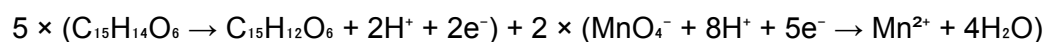


Figure 2 Mechanism of oxidation of catechin (Chattopadhyay et al.)

When there is abundant catechin, KMnO_4 will be reduced from purple (Mn^{7+}) to colorless (Mn^{2+}). The endpoint of titration is observed when all catechins have reacted and the solution. The more catechins present in the solution, the more potassium permanganate will be required to reach the endpoint.

Hypothesis: Studies have shown that catechins are more stable in acidic aqueous solutions. In solutions with pH above 6, catechins are unstable and degrade rapidly due to oxidation (Ahmad et al.). This means fewer catechins are available in the solution to react with the titrant. Therefore, as the pH of the brewing solution increases, the concentration of catechins in the tea solution will decrease.

Methodology:

Variables:

Table 1. Independent and Dependent variables

	Description	Scope/quantity	Manipulation/ measurement
Independent variable	pH of the tea solution	• 5 levels: pH 2, 4, 6, 8, 10 (chosen to cover a range from acidic to basic conditions) • An interval of 2 is adequate for recognizing a trend without unnecessary repetition of trials which waste time and resources.	• Adjusted using buffer capsules with uncertainty of ± 0.02 • Buffer capsules are chosen because they are more precise, stable and safe than strong acids and bases.
Dependent variable	Catechin concentration	—	• Determined indirectly via redox titration and stoichiometry calculations.

Table 2. Controlled variables

Controlled Variables	Reason for control	Method of Control
Mass of tea leaves	The amount of tea leaves used directly affects the concentration of catechins extracted.	0.5 g of tea leaves is used for each trial, weighed using an electronic balance ($\pm 0.01\text{g}$).
Volume of buffer solution	If different volumes were used, the concentration of catechins would be affected.	The volume for every trial is 25mL, measured using a graduated cylinder ($\pm 0.5\text{mL}$).
Steeping temperature	Higher temperatures extract more catechins, while lower temperatures extract less (Perva-Uzunalić et al.). Therefore the temperature should be consistent across trials so the observed variation in catechin concentration is mainly caused by change in pH not temperature.	The buffer solution is heated to 80°C for each trial, a typical brewing temperature for green tea(KitchenAid), and the temperature is monitored with a thermometer($\pm 1^{\circ}\text{C}$).
Steeping time	More catechins will be dissolved with longer steeping time. If steeping time varies, the catechin content would not be solely dependent on pH.	A stopwatch ($\pm 0.3\text{s}$) is used to time 10 seconds for all trials
Tea leaf source and type	Different tea brands and batches may have different catechin content due to different growing, processing and storing conditions.	The same brand and same batch of green tea leaves is used for all trials.
Concentration of KMnO_4	KMnO_4 is the titrant that reacts with catechins. The titration result is only reliable when the concentration of titration is consistent.	A standardized 0.05 M KMnO_4 solution is prepared and used.
Volume and concentration of sulfuric acid	Sulfuric acid is added to acidify the solution in order for a proper redox reaction to take place. If the volume or concentration changes, the reaction conditions could change, affecting the titration results.	A graduated cylinder ($\pm 0.5\text{mL}$) is used to measure 10 mL of 1M H_2SO_4 for each titration. All sulfuric acid is laboratory grade and is obtained from the same container.
Source of water	Impurity of water could introduce other substances that interfere with the reaction.	Distilled water is used throughout the experiment to ensure purity and consistency.

Preliminary trials :

Preliminary trials are performed and adjustments are made to the procedure according to the feedback from the trials:

1. Due to variability in factors such as temperature, steeping time, tea brand and type, and the ratio of tea to water, it was difficult to find studies conducted under the same conditions. Therefore, there is no reference for determining the appropriate concentration of KMnO_4 as the titrant. If the concentration is too low, a large volume of titrant will be needed before reaching the endpoint. If the concentration is too high, the endpoint may be reached too quickly, which increases the difficulty of endpoint determination and affects the accuracy of the result. I tested multiple concentrations ranging from 0.02 M to 0.1 M and found that 0.05 M was optimal. This concentration ensures sufficient volume of titrant to be added to reach the endpoint, allowing for more precise observation.
2. Initially, traditional gravity filtration was used to remove solid tea leaves. However, this method took too long, which increased the steeping time as the tea leaves remained in contact with the solution for an inconsistent duration. This directly impacts the result as the steeping time has a significant effect on the content of catechin. To improve this, suction filtration was used to reduce the filtration time to approximately three seconds. This minimizes the time of additional steeping and maintains consistency across trials.
3. The typical ratio of green tea in Chinese culture is 150ml of water with 3 g of tea (Yu). After multiple trials, the ratio is preserved to mimic the traditional brewing technique while being scaled down to 25 mL of water with 0.5g of tea to match the scale of the study.

Materials and Apparatus:

Materials:

- pH Buffer Capsules (pH = 2,4,6,8,10 $\pm$ 0.02)
- Green tea leaves
- Distilled water
- 1 M Sulfuric Acid Solution (H_2SO_4)
- Potassium Permanganate Solid (KMnO_4)

Table 3. Apparatus

Apparatus	Size/Uncertainty
Electronic Balance	$\pm 0.01\text{g}$
Hot Plate	-
Thermometer	$\pm 1\text{ }^{\circ}\text{C}$
Beaker (for boiling/ dissolving solution)	250mL
Filter Paper	-
Suction Filtration Apparatus	-
Volumetric Flask	250 $\pm 0.15\text{ mL}$
Graduated cylinder	$\pm 0.5\text{ mL}$
Weighing Boat	-
Funnel	-
Stirring Rod	-
Erlenmeyer Flask	250mL
Burette	$\pm 0.05\text{ mL}$
Ring stand	-
Buret clamp	-
Stopwatch	$\pm 0.3\text{ s}$ according to human reaction time(Human Benchmark)
Safety goggles	-
Lab coat	-

Procedure:

I. Preparation of KMnO_4 Solution:

Calculate the mass of potassium permanganate solid required to produce 250 ml of 0.05 M solution:

$$n = M \times V$$

M is the molarity (in moles per liter, M)

n is the number of moles

V is the volume of solution (in liters)

Plug in the formula:

$$M = 0.05\text{M}$$

$$V = 250\text{mL} = 0.25\text{L}$$

$$n = 0.05\text{M} \times 0.250\text{L} = 0.0125 \text{ moles}$$

Calculate the molar mass of potassium permanganate (KMnO_4):

Potassium (K) = 39.1 g/mol,

Manganese (Mn) = 54.9 g/mol,

Oxygen (O) = 16.0 g/mol

(PubChem)

$$39.1 + 54.9 + (4 \times 16.0) = 39.1 + 54.9 + 64.0 = 158.0\text{g/mol}$$

Calculate the mass of KMnO_4 required:

$$\begin{aligned}\text{Mass} &= n \times \text{molar mass} \\ &= 0.0125 \text{ moles} \times 158.0 \text{ g/mol} \\ &= 1.975\text{g} \\ &\approx 1.98\text{g}\end{aligned}$$

1. Weight 1.98 g of potassium permanganate using electronic balance and weighing boat
2. Transfer the potassium permanganate in a beaker and dissolve with a small amount of distilled water.
3. Rinsed the weighing boat to ensure all potassium permanganate is transferred to the beaker

4. Transfer the solution into the volumetric flask using a funnel
5. Rinsed the beaker, stirring rod, and funnel to ensure complete transfer of potassium permanganate.
6. Slowly add distilled water into the volumetric flask until the bottom of the meniscus is level with the graduation mark.
7. This will produce a 0.05 M solution, which is the titrant in this experiment.

II. Preparation of Green Tea Solution of different pH:

1. Weigh 0.5g of green tea leaves using an electronic balance
2. Prepare solutions of the desired pH using the buffer capsule (pH = 2,4,6,8,10)
3. Boil the 25 ml of the solution to 80°C with a hot plate. Monitor the temperature using the thermometer.
4. Pour the heated solution over the tea leaves and swirl for 10 seconds.
5. Filter the tea extract using the suction filtration method to remove solid tea leaves
6. Allow the filtrate to cool to room temperature before titration.
7. Repeat the process for solutions of different pH.

III. Titration Procedure:

1. Pour 25 ml of the tea solution into an erlenmeyer flask.
2. Add 10 mL of 1 M sulfuric acid to acidify the solution
3. Fill the burette with 0.05 M potassium permanganate solution and record the initial volume.
4. Slowly titrate the tea solution while swirling the flask.
5. Continue adding KMnO_4 until a light pink color remains for 30 seconds, indicating the endpoint.
6. Record the final volume of KMnO_4 .
7. Repeat titration 3 times for each and every pH as well as the control group using distilled water

Safety, Environmental, and Ethical Considerations :

Safety Considerations

- **Handling Chemicals:**

Potassium permanganate (KMnO_4), as an oxidizing agent, can cause skin and eye irritation. Sulfuric acid (H_2SO_4) is highly corrosive and can cause severe burns. Thus, lab coats and safety goggles should be worn at all times. In case of contact, the affected area should be rinsed immediately with water.

Environmental Considerations

- **Waste Disposal:**

- Potassium permanganate solution is a strong oxidizer that leads to environmental contamination and can be harmful to aquatic life(King). It should be disposed of in the designated waste container to avoid entering the local drainage system.
- Acidic and buffer solutions should be diluted before disposal down the sink to reduce environmental impact.

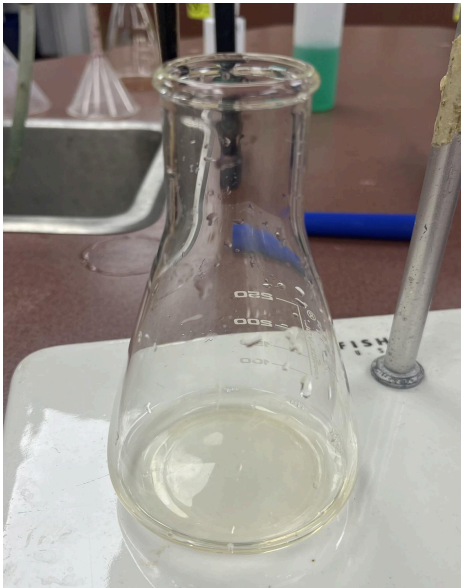
Ethical Considerations

- There are no ethical concerns regarding the procedure of this experiment as no living organism is involved and causes no harm to local communities.

Raw data:

Qualitative:

After reaching the endpoint, the color of the solution turned from colorless to light pink

Before endpoint	Reached endpoint
	

Quantitative :

Table 4: Raw Data of Titration for Tea Brewed with pH 2 Buffer Solution

pH 2		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	33.00	34.85
2	30.00	33.50
3	41.00	43.05

The tables of titration raw data for pH 4,6,8,10 solutions and control are included in the appendix.

Data processing

I. Calculate the concentration of catechin (using pH 2 as an example)

Main calculations	Calculating uncertainty
Volume of KMnO_4 used Final - Initial = Volume used $34.85 - 33 = 1.85$ mL $33.9 - 30 = 3.50$ mL $43.05 - 41 = 2.05$ mL Average calculations (KMnO_4) Volume $\frac{1.85 + 3.50 + 2.05}{3} = \frac{7.40}{3}$ $= 2.47$ mL $2.47 \text{ mL} \rightarrow 0.00247\text{L}$ $0.00247\text{L} \pm 4.05\%$ Calculate the Moles of KMnO_4 Used: $n = M \times V$ $n = 0.05 \times 0.00247$ $= 1.235 \times 10^{-4}$ moles	$0.05 + 0.05 = 0.1\text{mL}$ $0.1 + 0.1 + 0.1 = 0.3\text{mL}$ $= 7.40 \pm 0.3\text{mL}$ $= 7.40 \text{ mL} \pm 4.05\%$

Calculate the Moles of Catechin in the Tea Sample: From the balanced reaction: $5\text{C}_{15}\text{H}_{14}\text{O}_6 + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow 5\text{C}_{15}\text{H}_{12}\text{O}_6 + 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$ The mole ratio of catechin to permanganate is 5:2. $1.235 \times 10^{-4} \times \frac{5 \text{ catechin}}{2 \text{KMnO}_4} = 3.0875 \times 10^{-4} \text{ moles}$ Calculate the Concentration of Catechin in the Tea Solution Volume of tea solutions: 25.0 mL $\rightarrow$ 0.025L $C = \frac{n}{V}$ $= \frac{3.0875 \times 10^{-4}}{0.025}$ $= 0.01235 \text{ M}$ $= 0.01235 \pm 6.05\%$ $= \mathbf{0.0124 \pm 0.0007 \text{ M}}$	25.0 $\pm$ 0.5mL = 25.0 $\pm$ 2% mL 2% + 4.05% = 6.05%
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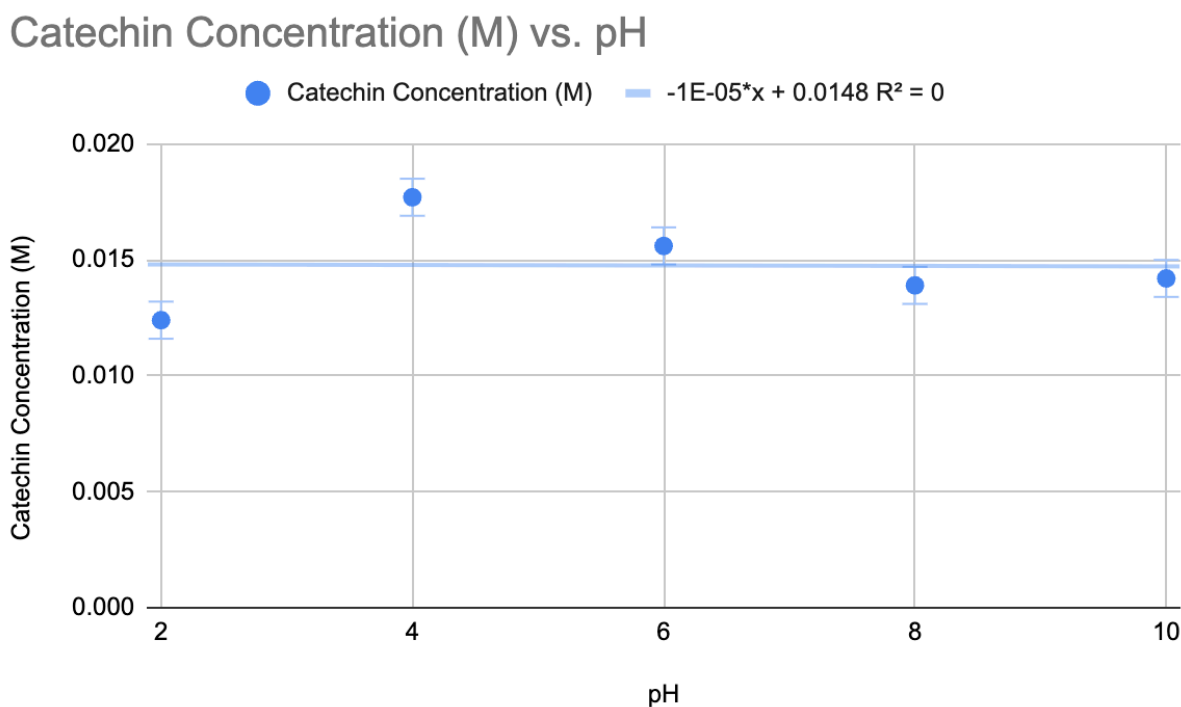
Using the same method as above, the concentrations and uncertainties of different pH trials and control group is calculated, see below:

Table 5. Concentration of Catechin in Tea Solutions at Different pH Levels with Uncertainty

pH	Concentration of Catechin with (M)	percentage uncertainty
2	0.0124 $\pm$ 0.0007 M	6.05%
4	0.0177 $\pm$ 0.0008 M	4.80%
6	0.0156 $\pm$ 0.0008 M	5.21%
8	0.0139 $\pm$ 0.0008M	5.60%
10	0.0142 $\pm$ 0.0008M	5.52%
Control (distilled water)	0.0159 $\pm$ 0.0008 M	5.14%

The percentage uncertainty of data ranges from 4.80%-6.05%, which are relatively small. This means the measurements made in this experiment are reliable and precise. Thus, the variations introduced by apparatus do not have a significant impact on the result of this study.

Figure 1. Catechin Concentration (M) vs pH



Evaluation and Conclusion:

The aim of this experiment was to investigate the effect of pH on the catechin concentration in the context of brewing green tea. Redox titration with potassium permanganate as titrant is used to determine the concentration of catechin under different pH conditions. The result (shown in Figure 1) indicates no correlation between pH and concentration of catechin. The very flat trendline with slope close to 0 (-1×10^{-5}) suggests that pH has little to no effect on the catechin concentration. Since the control group's value aligns with the other trials, it further supports that pH does not influence catechin concentration. This finding rejects my initial hypothesis, stating the catechin concentration will decrease as pH increases. The theory behind the hypothesis is that catechins degrade at high pH because alkaline conditions promote rapid oxidation (Zeng et al.). While this holds true, further studies have shown that the duration of the degradation process of catechin is longer than expected. A study observed that 80 % catechin is lost in the sodium phosphate buffer of (pH 7.4) after 3 hours (Zeng et al.). However, in this experiment, titration is performed within 10 minutes after brewing. This is a possible explanation of why pH has no effect on the catechin content as the degradation of catechin occurs over time.

Limitation as Weakness:

Moreover, even though the error bars are small, this only reflects the precision of the apparatus. Other unquantifiable factors in the experimental process also have a significant impact on the precision and accuracy of the result.

Limitations/Weakness	Effect on the result	Possible improvements
Detection of Endpoint (Random error)	The visual detection of the endpoint is subjective as people have different sensitivity to color. This results in a large probability of human error and inconsistency in the catechin concentration calculations.	Colorimeters could be used to determine the color change more precisely and objectively.
Other substances that can oxidize (systematic error)	Even though Catechin is the most abundant and active reducing agent, other substances such as Vitamin C or other types of polyphenol also undergoes oxidation(Sun et al.). This leads to overestimating the catechin concentration in all trials	UV-Vis Spectroscopy can test for a specific polyphenol by using a selected wavelength, allowing the determination of catechins content without interference from other polyphenols or Vitamin C Or If equipment like UV-Vis spectrophotometer is not available, a controlled trial could be run with pure catechin of known concentration. By comparing the experimental value to this standard value, the difference provides insight into how much overestimation is caused by other compounds.
Inconsistent Swirling Technique (Random error)	The swirling motion is difficult to maintain the same across trials, leading to variations in the extraction efficiency of catechins. If swirled more vigorously, the more catechins may be extracted.	Using a stirring hot plate can maintain uniform mixing motion which eliminates variation caused by manually stirring.

Loss of Tea Solution in Filtration (systematic error)	Some tea solution remains on the filter paper, which means the volume used in titration is less than the intended 25 mL. Since a larger volume than the actual amount of solution is used, the result will be a lower concentration than the true value.	Remeasure the volume of the tea solution after the filtration to determine the actual volume. Or Measure the mass of the filter paper before and after the filtration. The difference in mass is equivalent to the mass of tea solution lost. Since the tea solution is mostly water, the 1 g to 1 mL conversion could be used to determine the amount of water lost. This allows the actual amount of water used in titration to be accurately determined.
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Improvements and further investigation:

As mentioned above, the challenges regarding the endpoint determination greatly influenced the result of this study. Additionally, the redox reaction in this experiment is relatively slow, as the color may initially change completely but slowly revert to colorless within 30 seconds. This makes it extremely difficult to accurately time the 30-second period when the color remains changed. Further research has also proven that the titration method for determining polyphenol content is less accurate and has lower sensitivity compared to other methods(Sun et al.) Therefore, spectrophotometric monitoring of catechin degradation may be a better option. The yellow color of the tea correlates with oxidation, which can be tracked through these absorbance values(Sun et al.). Furthermore, the absorbance can be monitored over a longer period of time, for example 24 hours, which provides valuable data into the rate of change in catechin content at different pH levels.

This can also be carried forward into the investigation of how other factors such as temperature, humidity and light exposure affect catechin degradation(Li et al.). In the process of tea processing, storage, and brewing, catechins are lost due to breakdown with time, reducing the health benefits of tea. With increased understanding of the ways these conditions impact the catechin degradation, researchers are able to decide upon optimal processing strategies and storage routines that ensure maximal amount of active catechins and complete health effects of tea.

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Appendix:

Titration Raw data:

pH 4		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	34.00	36.85
2	36.00	38.95
3	42.00	46.80

pH 6		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	40.00	43.35
2	32.00	36.15
3	34.00	35.85

pH 8		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	44.00	46.45
2	37.00	40.35
3	40.00	42.55

pH 10		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	40.00	41.55
2	39.00	41.85
3	37.00	41.12

Distilled water		
Trial	Initial Volume (± 0.05 mL)	Final Volume (± 0.05 mL)
1	35.00	38.75
2	38.00	41.45
3	42.00	44.35